

August 2025

Homework Problems

1. Introduction to Complex Numbers

1. Let $z = 3 - 5j$. Identify the real and imaginary parts of z . Plot it on the Argand diagram.
2. Given $z_1 = -2 + j$ and $z_2 = 4 - 3j$, determine whether $\Re(z_1 + z_2) = \Re(z_1) + \Re(z_2)$.
3. Write the general form of a complex number and define j . What is j^4 ?
4. If $z = a + bj$, show that $z + z^* = 2a$.
5. Plot the complex numbers $1 + j$, $-2 + 2j$, and $-1 - j$ on an Argand diagram.
6. Which of the following numbers are purely imaginary: $0 + 3j$, 2 , $-j$, $j - 1$?
7. Prove that the sum of a complex number and its conjugate is always real.

2. Rectangular Form Operations

8. Compute $(5 + j)(-2 + 4j)$ and express the result in rectangular form.
9. Simplify $\frac{3-2j}{1+j}$ and write your answer in the form $a + bj$.
10. If $z = 7 + 24j$, compute $|z|$ and verify $|z|^2 = z \cdot z^*$.
11. Given $z = x + yj$, find expressions for $\Re(z^2)$ and $\Im(z^2)$.
12. Show that multiplication by j rotates a complex number by 90° counter-clockwise.
13. Let $z_1 = 3 + j$ and $z_2 = 1 - 2j$. Compute $\frac{z_1}{z_2}$.
14. Compute $(1 - j)^2$ and interpret geometrically on the Argand diagram.

3. Magnitude and Argument (Polar Form)

15. Express $-5 + 5j$ in polar form using magnitude and argument in degrees.
16. Convert the polar form $z = 10\angle 60^\circ$ to rectangular form.
17. If $z = 2\angle 45^\circ$, compute z^3 in polar form.
18. Find the magnitude and angle of $z = -6 - 8j$, and plot it.
19. Express $\sqrt{3} + j$ in polar form. Then calculate z^6 .
20. Prove that the conjugate of a complex number $z = r\angle\theta$ is $\bar{z} = r\angle(-\theta)$.
Hint: Recall that $\cos(-\theta) = \cos(\theta)$ and $\sin(-\theta) = -\sin(\theta)$.

Solutions

1. Introduction to Complex Numbers

1. $\Re(z) = 3$, $\Im(z) = -5$. Plotted at point (3, -5) on Argand diagram.
2. $z_1 + z_2 = 2 - 2j$, so $\Re(z_1 + z_2) = 2 = \Re(z_1) + \Re(z_2)$. True.
3. $z = a + bj$, where $j^2 = -1$. Thus, $j^4 = 1$.
4. $z + z^* = (a + bj) + (a - bj) = 2a$, which is real.
5. Plot:
 - $1 + j$: (1, 1)
 - $-2 + 2j$: (-2, 2)
 - $-1 - j$: (-1, -1)
6. Purely imaginary: $0 + 3j$, $-j$. Others are not.
7. Let $z = a + bj$, then $z^* = a - bj$, and $z + z^* = 2a$, which is real.

2. Rectangular Form Operations

8. $(5 + j)(-2 + 4j) = -10 + 20j - 2j + 4j^2 = -10 + 18j - 4 = -14 + 18j$
9. Multiply numerator and denominator by $1 - j$:
$$\frac{3 - 2j}{1 + j} \cdot \frac{1 - j}{1 - j} = \frac{(3 - 2j)(1 - j)}{1^2 + 1^2} = \frac{3 - 3j - 2j + 2j^2}{2} = \frac{3 - 5j - 2}{2} = \frac{1 - 5j}{2} = 0.5 - 2.5j$$
10. $|z| = \sqrt{7^2 + 24^2} = \sqrt{625} = 25$, and $|z|^2 = 625 = z \cdot z^*$
11. $z^2 = (x + yj)^2 = x^2 + 2xyj + y^2j^2 = x^2 - y^2 + 2xyj$, so:
$$\Re(z^2) = x^2 - y^2, \quad \Im(z^2) = 2xy$$
12. Let $z = a + bj$. Then:
$$jz = j(a + bj) = aj + bj^2 = aj - b$$
So $jz = -b + aj$: rotates counterclockwise by 90° .
13. Multiply numerator and denominator by conjugate:
$$\frac{3 + j}{1 - 2j} = \frac{(3 + j)(1 + 2j)}{1^2 + 4} = \frac{3 + 6j + j + 2j^2}{5} = \frac{3 + 7j - 2}{5} = \frac{1 + 7j}{5} = 0.2 + 1.4j$$
14. $(1 - j)^2 = 1 - 2j + j^2 = 1 - 2j - 1 = -2j$, a pure rotation to the imaginary axis.

3. Magnitude and Argument (Polar Form)

15. $r = \sqrt{(-5)^2 + 5^2} = \sqrt{50} = 5\sqrt{2}$, $\theta = \tan^{-1}(-1) = -45^\circ$, but adjust for quadrant: 135° , so $z = 5\sqrt{2}\angle 135^\circ$

16. $z = 10\angle 60^\circ = 10(\cos 60^\circ + j \sin 60^\circ) = 10\left(\frac{1}{2} + j\frac{\sqrt{3}}{2}\right) = 5 + 5\sqrt{3}j$

17. $z^3 = 2^3\angle 3 \times 45^\circ = 8\angle 135^\circ$

18. $r = \sqrt{36 + 64} = 10$, $\theta = \tan^{-1}(8/6) \approx 53.13^\circ$, but 3rd quadrant $\rightarrow -126.87^\circ$ or 233.13°

19. $r = \sqrt{3 + 1} = 2$, $\theta = \tan^{-1}(1/\sqrt{3}) = 30^\circ$, so $z = 2\angle 30^\circ$, Then $z^6 = 64\angle 180^\circ = -64$

20. Conjugate of $z = r\angle\theta$:

$$z = r(\cos \theta + j \sin \theta) \quad \Rightarrow \quad \bar{z} = r(\cos \theta - j \sin \theta)$$

$$= r(\cos(-\theta) + j \sin(-\theta)) = r\angle(-\theta)$$

$$\bar{z} = r\angle(-\theta)$$

21. Conjugate of $z = a + jb$ where $a = r \cos \theta$, $b = r \sin \theta$:

$$z = a + jb \quad \Rightarrow \quad \bar{z} = a - jb$$

$$\text{Then, } |z| = \sqrt{a^2 + b^2}, \quad \arg(z) = \tan^{-1}\left(\frac{b}{a}\right)$$

$$\Rightarrow \quad \arg(\bar{z}) = \tan^{-1}\left(\frac{-b}{a}\right) = -\arg(z)$$

$$\bar{z} = r\angle(-\theta)$$